Mid-Internship Review

Ziheng Wang · Data Analyst Intern · inGen Dynamics (Futurenauts) · Weeks 1–8 · prepared for supervisor review

Summary

Through Week 8 I completed Phase 1 (industry & data foundation) and Phase 2 (market & competitive analytics), and built out Phase 3 (BI development). The work is a connected arc: understand the five verticals, engineer a public-data foundation, size the markets, read demand and competitor financials, then stand up a queryable warehouse and dashboards on top. A consistent discipline runs through all of it — real sourced data, ranges where there's genuine uncertainty, and clearly-labelled fallbacks where a live source wasn't reachable.

Phase status

Phase	Weeks	Status
1 — Industry & data foundation	1–3	Complete
2 — Market & competitive analytics	4–6	Complete
3 — BI development & visualization	7–9	Wk7 complete; Wk8 built (publish pending); Wk9 in progress
4 — Advanced analytics	10–13	Upcoming

(a) What shipped — Weeks 1–8

Wk	Deliverable	Artifact
1	Product profiles, competitor landscape (40+), data dictionary	week1 repo
2	15 priority peers: real patents, headcounts, IP-vs-funding analysis	week2 repo
3	12-dataset ingestion pipeline, DuckDB warehouse, 15 tests, DQ report	week3 repo
4	TAM/SAM/SOM all 5 verticals, dual-method, tornado sensitivity, workbook	week4 repo
5	Demand signals (search/news/VoC) + demand-signal index; pain-point taxonomy	week5 repo
6	Peer financial benchmark (public FY24 + private funding), one-pager, memo	week6 repo
7	Star-schema warehouse, 100k-row synthetic data, 15 queries, SQL self-assessment 12/12	week7 repo
8	Two dashboards (Tableau + Looker): specs, real extracts, prototypes, design log	week8 repo

Every week is a self-contained repo with a README and tests where applicable; all are linked from the repository INDEX.md.

(b) What worked · (c) What slowed progress

What worked

- **Reusable foundation paid off.** The Week 3 ingestion pipeline + DuckDB warehouse were reused directly in Weeks 4, 7, and 8 — building it once accelerated everything after.
- **Data-integrity discipline.** Insisting on sourced values, ranges over false precision, and labelled fallbacks made the work defensible — e.g., the Week 6 benchmark cites a source per row and flags estimated cells.
- **Cross-week synthesis.** Findings compounded: Week 2's “funding ≠ defensible IP” was confirmed by Week 6's financials; Week 4 sizing + Week 5 demand together give a prioritization the company can act on.
- **Testing.** pytest suites (Weeks 3, 5, 6, 7) caught regressions and make every deliverable reproducible.

What slowed progress

- **Network-restricted environment.** Live sources (Google Trends, GDELT, SEC EDGAR) aren't reachable from the build sandbox, so several weeks ship real code against real APIs plus labelled synthetic fallbacks. Fully populating live data needs a networked run.
- **External BI tools are manual.** Tableau Public, Looker Studio, and Power BI publish from my own accounts; I built specs/extracts/prototypes, but the final publish + URLs are a manual step.
- **No internal InGen data.** Market bottom-ups and product analytics rest on public/synthetic data; internal cost/pilot/CRM data would move several outputs from “directional” to “actionable.”

(d) Phase 4 goals (Weeks 10–13)

Phase 4 applies advanced analytics to InGen use cases. Concrete goals:

- **Week 10 — Forecasting.** Turn the Week 5 demand signals into supervised targets; fit baselines (seasonal naive, ETS, ARIMA) then Prophet + XGBoost with engineered features; Bass diffusion for humanoid adoption. Compare on MAPE/MASE; sanity-check vs published analyst forecasts.
- **Weeks 11–13 — applied modeling + capstone.** Extend into the remaining Phase 4 use cases and consolidate the internship into a capstone narrative that ties market → demand → financials → forecasts.
- **Standing objectives.** Keep every deliverable reproducible and sourced; convert at least one pipeline to a live networked run; and, where possible, validate against any internal data made available.

Success measure for Phase 4

Forecasts that beat naive baselines on MAPE/MASE, are explainable by named drivers, and are honestly bounded with confidence intervals — not single-point numbers.

(e) One explicit ask of my supervisor

The ask	Can I get read access to one slice of real InGen data — even a small, anonymized export of fleet telemetry OR sales-pipeline records — so I can validate the Week 7 warehouse design and the Week 10 forecasts against reality? The synthetic warehouse mirrors the intended structure, so a real export would slot in with minimal rework and would move the analytics from directional to decision-grade.

Why this is the highest-leverage ask

Most of what currently carries a caveat — bottom-up market assumptions, demand-as-proxy, synthetic product analytics — resolves the moment any real internal data is available to calibrate against. One representative export unlocks disproportionate value across Phases 3 and 4.

Secondary (smaller) asks

- A networked environment (or approval to run locally) to populate the live Google Trends / GDELT / SEC pulls.
- 15 minutes to confirm the five product names/positioning and any KPI targets the leadership team actually uses, so the scorecard reflects real targets.

Prepared by Ziheng Wang · inGen Data Analyst internship · mid-internship self-review · Weeks 1–8 shipped, Phase 4 next. All artifacts linked from the repository INDEX.md.